

Clinical Early Warning System

Dataset: PhysioNet Sepsis Challenge 2019

Section 1: Problem Statement Why This Is Hard and Why Recall Matters Most

Sepsis kills 1 in 5 ICU patients. Every hour of delayed treatment raises mortality by 7%. An AI system that predicts sepsis 6 hours early can directly save lives. Our dataset contains 1,552,210 hourly ICU readings but only 1.8% are sepsis cases. This severe class imbalance is the central challenge of the entire project.

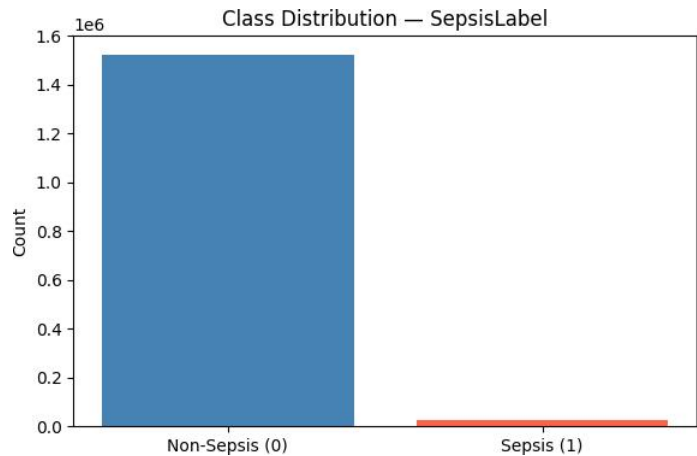


Figure 1: Class Distribution

What this chart tells us: The red bar is barely visible compared to the blue bar. A model that predicts "no sepsis" for every patient still scores 98.2% accuracy yet it catches zero real cases. This is the accuracy trap in imbalanced medical data, and it is the core reason why Recall, not Accuracy, drives all decisions in this project.

Recall measures: out of all real sepsis patients, how many did the model correctly flag? Missing a sepsis patient (false negative) can be fatal. A false alarm (false positive) only causes extra monitoring inconvenient but not deadly. Every architecture decision, class weight, and optimizer choice in this project was made to maximize Recall, even when it reduces precision or accuracy.

Metric	What It Measures	Clinical Importance
Recall	Sepsis cases correctly identified	MOST CRITICAL missed = patient at risk
Precision	Correct among all raised alarms	Secondary false alarms cost resources
Accuracy	Overall correct predictions	MISLEADING with 98/2 class imbalance
F1 Score	Balance of precision and recall	Useful overall summary

Table 1: Why Recall is the primary evaluation metric

Sepsis also develops gradually a rising heart rate trend over 12 hours predicts sepsis far better than any single reading. Flat models miss this. This is the core reason for progressing through three model generations.

Section 2: Generation 1 Deep Neural Network Design

Feature Selection 3-Step Justification

Starting from 43 raw columns, features were selected using three independent methods. Any column with more than 70% missing values was dropped first 27 columns including Bilirubin, Fibrinogen, Lactate, and Troponin were almost never recorded in practice. Then Random Forest importance scored each remaining feature. Finally, Point-Biserial correlation confirmed statistical significance. Gender, Unit1, and Unit2 were removed near-zero RF score and weak correlation. Final 11 features kept: Hour, ICULOS, HR, Resp, Temp, SBP, MAP, DBP, Age, HospAdmTime, O2Sat.

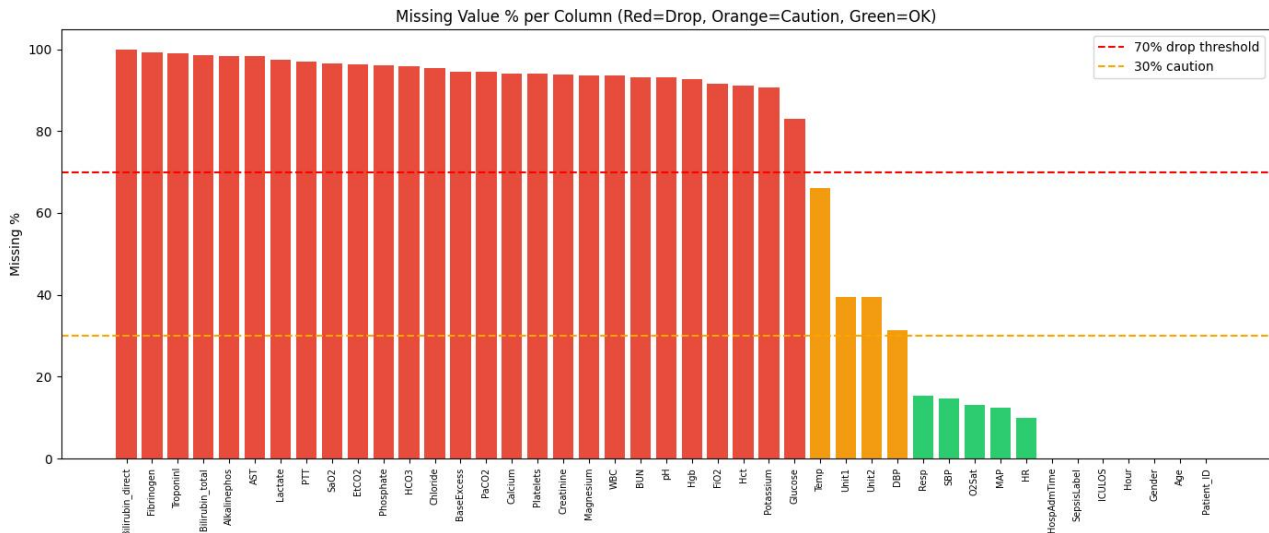


Figure 2: Missing Value % per Column (Red=Dropped >70%, Orange=Caution, Green=Reliable)

What this chart tells us: The 27 red bars on the left were all removed. Bilirubin_direct had 99.8% missing, Fibrinogen 99.3%, Lactate 97.3%. These columns existed but were almost never measured keeping them would add noise. The green bars on the right (HR, O2Sat, SBP, MAP, Resp) had the least missing data and are the most clinically reliable vital signs exactly what we kept.

Figure 3: Random Forest Feature Importance

ICULOS (0.235) is strongest longer ICU stay = higher sepsis risk. HospAdmTime, Age, HR, SBP, MAP follow. Gender/Unit1/Unit2 are grey near-zero bars dropped.

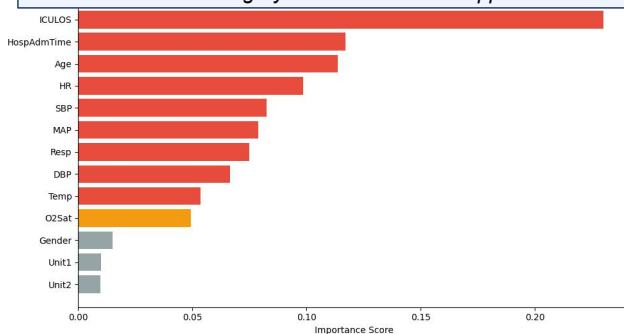
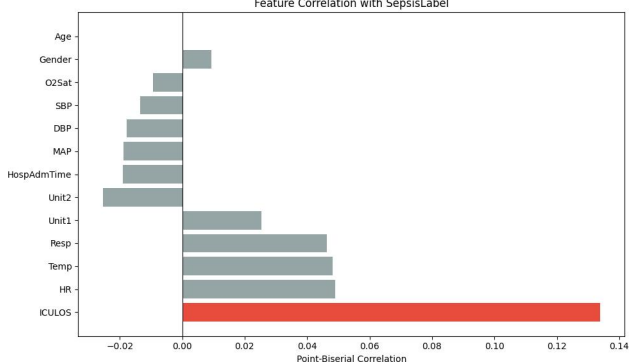


Figure 4: Point-Biserial Correlation with SepsisLabel

ICULOS (0.133) strongest correlation. HR, Temp, Resp show positive correlations classic Sepsis-3 clinical signs. SBP.



DNN Architecture, Optimizers, and Training Results

Three Dense layers (128→64→32 neurons). Each layer followed by Batch Normalization (normalizes activations, stabilizes gradients, allows higher learning rates) and Dropout (0.3/0.3/0.2 randomly disables neurons to prevent overfitting). Sigmoid output for binary classification. Class weights 27.8x for sepsis every missed sepsis case penalized 27.8x harder, directly training the model to prioritize Recall over accuracy. SGD (lr=0.01, momentum=0.9) vs Adam (lr=0.001) compared across 50 epochs with early stopping.

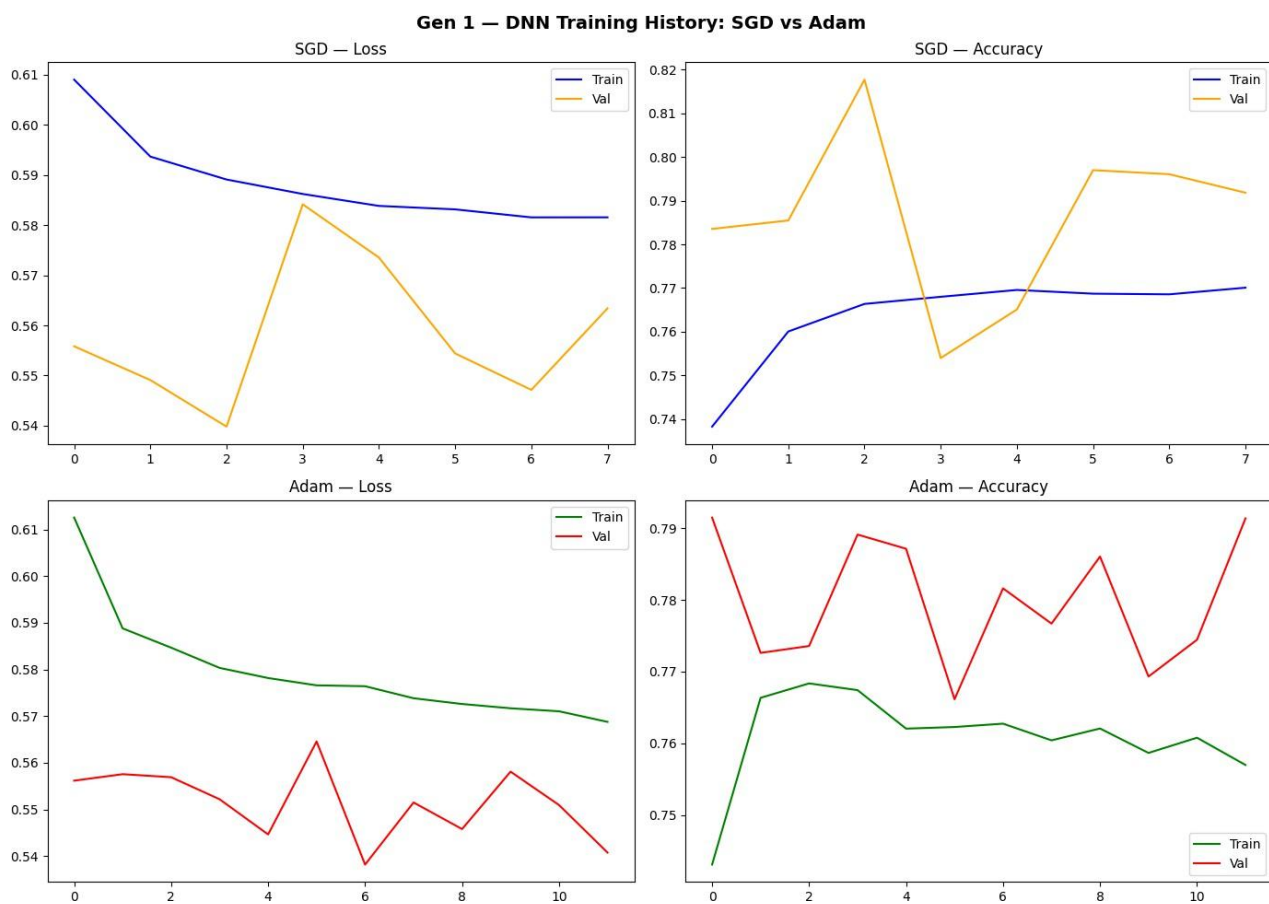
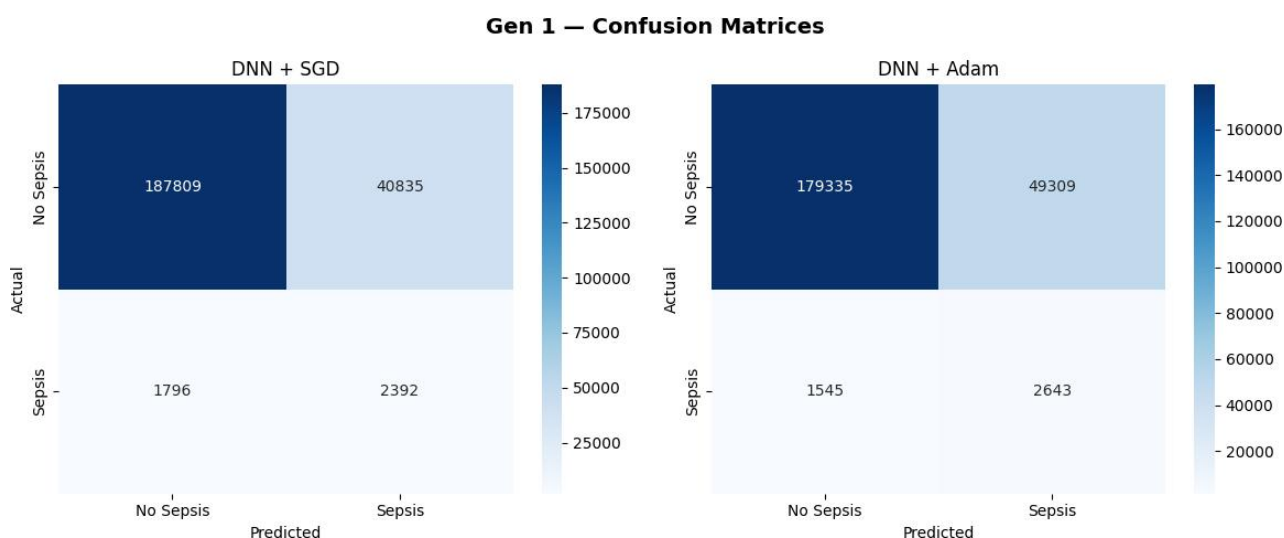


Figure 5: Gen 1 Training History SGD (top row) vs Adam (bottom row)

What these curves tell us: SGD (blue/orange) validation loss oscillates heavily, accuracy jumps between 75–82%, showing SGD struggles with this imbalanced data. Adam (green/red) converges faster, val and train curves stay closer together, showing better generalization. Adam triggered early stopping at epoch 12 vs SGD at epoch 8, meaning Adam kept improving for longer.

Figure 6: Gen 1 Confusion Matrices DNN+SGD (left) vs DNN+Adam (right)



What these matrices tell us: Bottom-right = true positives (sepsis correctly caught). Bottom-left = missed sepsis cases (false negatives the most dangerous error). SGD caught 2,392 sepsis cases, missed 1,796. Adam caught 2,643 sepsis cases, missed 1,545 catching 251 more patients. Adam Recall = 0.631 vs SGD 0.571. Adam is the better optimizer for this task.

Model	Accuracy	Precision	Recall	F1 Score	Train Time
DNN + SGD	0.8169	0.0553	0.5712	0.1009	59s
DNN + Adam	0.7816	0.0509	0.6311	0.0942	315s

Table 2: Gen 1 Results Adam wins on Recall (0.631 vs 0.571)

Recall Focus: Best Gen 1 Recall = 0.631 (Adam). Catches 63% of sepsis patients from flat tabular data. Missing 37% because the DNN treats each hourly reading independently it has no concept of trends or temporal patterns.

Section 3: Generation 2 LSTM, Bidirectional LSTM, and GRU

The DNN's core limitation: it treats every hourly reading independently with no memory. Gen 2 fixes this by reshaping data into patient sequences. All rows for each patient are grouped by Patient_ID, sorted by Hour, and the last 24 hours form one sequence (shape: 24 timesteps × 11 features). Patients with fewer than 24 hours of data are zero-padded at the beginning. Label = 1 if patient developed sepsis at any point. At the patient level, balance improves to 92.7% vs 7.3% class weight drops to 6.85x.

LSTM uses 3 gates (forget, input, output) to remember patterns across many hours (32,417 params). **GRU** merges gates into 2, reducing params to 24,737 24% fewer while matching LSTM performance, training 4x faster (22s vs 86s). **Bidirectional LSTM** runs two LSTMs forward AND backward (81,185 params), achieving the highest Recall of 0.726. However it cannot run in real time reading backwards requires future hourly data that does not exist yet in a live ICU. Bidirectional LSTM is suitable only for offline retrospective analysis.

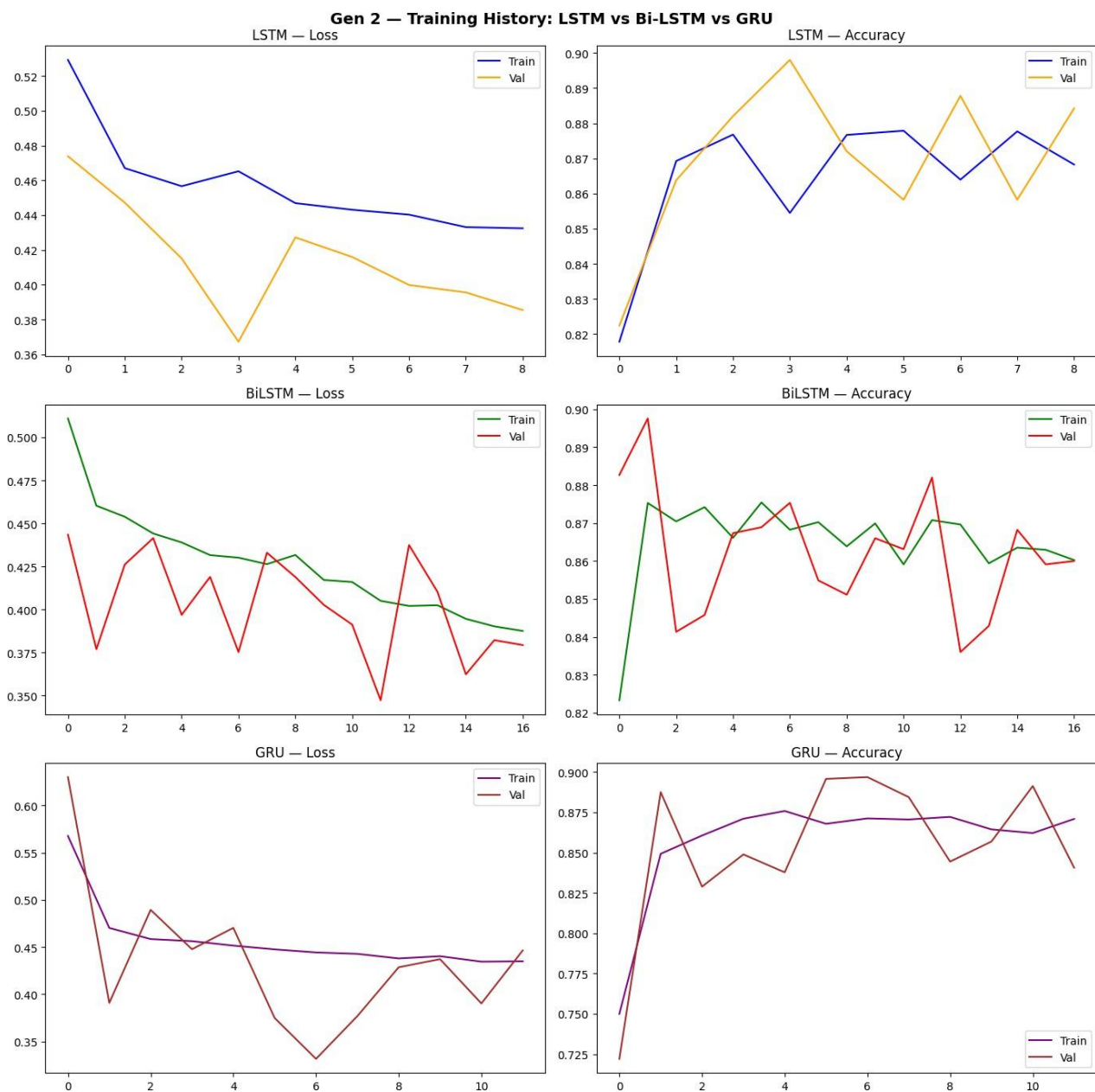


Figure 7: Gen 2 Training History LSTM (top), Bi-LSTM (middle), GRU (bottom)

What these curves tell us: GRU (purple/brown, bottom) has the healthiest training both loss curves decrease smoothly together, converging in 11 epochs with minimal oscillation. Bi-LSTM (green/red, middle) is most volatile across 17 epochs, the red validation line jumps repeatedly 2.5x more parameters need more epochs to settle. LSTM (blue/orange, top) converges cleanly in 9 epochs. All three reach 87-90% validation accuracy.

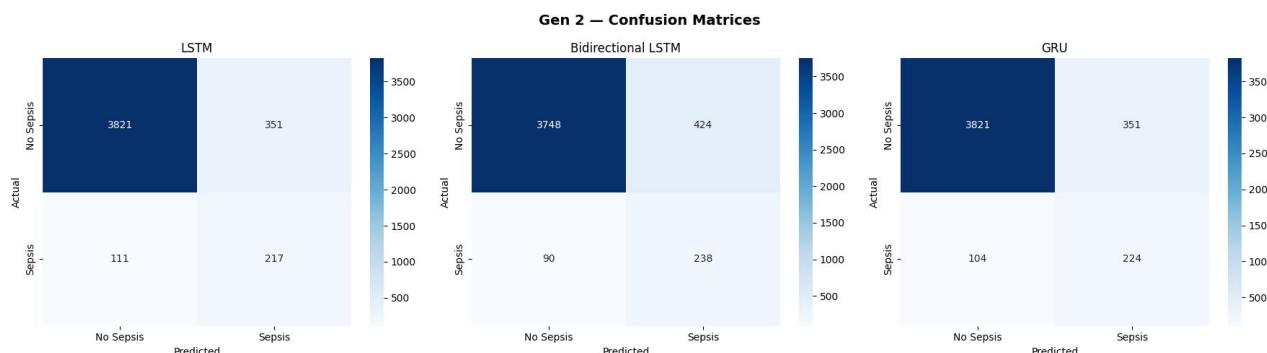


Figure 8: Gen 2 Confusion Matrices LSTM (left), Bidirectional LSTM (middle), GRU (right)

What these matrices tell us: Bottom-right value = true positives (sepsis caught). Bi-LSTM caught 238 sepsis patients the most. LSTM caught 217, GRU caught 224. Bi-LSTM missed only 90 patients vs LSTM missing 111. The difference of 21 extra patients caught could be life-saving in a real hospital.

Model	Accuracy	Precision	Recall	F1	Params	Train Time
LSTM	0.8973	0.3820	0.6616	0.4844	32,417	86s
Bidirectional LSTM	0.8858	0.3595	0.7256	0.4808	81,185	186s
GRU	0.8989	0.3896	0.6829	0.4961	24,737	22s

Table 3: Gen 2 Bi-LSTM best Recall (0.726); GRU best efficiency for deployment

Recall Focus: Best Gen 2 Recall = 0.726 (Bi-LSTM) crosses the 0.70 clinical target. For live real-time deployment: GRU Recall = 0.683 in just 22 seconds. Recall jumped from 0.631 (DNN) to 0.726 (+0.095) temporal trend modeling is the key driver of improvement.

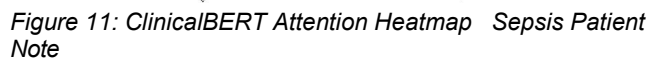
Section 4: Generation 3 ClinicalBERT, Attention Heatmap, Frozen vs Fine-Tune

Gen 3 uses language instead of numbers. Bio_ClinicalBERT was pre-trained on MIMIC-III real hospital notes it already understands tachycardia, hypotension, and clinical sepsis indicators. Since PhysioNet has no text data, synthetic notes were generated from vitals: HR=105 → 'Tachycardia noted, heart rate 105 bpm.' SBP=88 → 'Hypotension detected, systolic BP 88 mmHg.' 5,000 notes generated with zero data leakage (word 'sepsis' never appears in notes). **Frozen:** only 1,538 params train (classifier head) fast but Recall = 0.250, worst of all. **Full fine-tune:** all 108M params adapt Accuracy = 95.1%, Precision = 0.649 (best of all models), Recall = 0.500.

Model	Accuracy	Precision	Recall	F1 Score	Train Time
ClinicalBERT Frozen	0.8600	0.1481	0.2500	0.1860	1,326s
ClinicalBERT Full Fine-Tune	0.9507	0.6486	0.5000	0.5647	6,239s

Table 4: Gen 3 Full fine-tune best Accuracy (95%) and Precision (0.649)

Recall Focus: Best Gen 3 Recall = 0.500 (Full Fine-Tune). Lower than LSTM models because only 5,000 synthetic notes were used BERT needs real clinical text at scale to reach its potential. Frozen Recall = 0.250 confirms that locking BERT weights is completely unsuitable.



Section 5: Deployment Decision, Ethics, and Multimodal Extension

Final Comparison All 6 Models

Model	Accuracy	Precision	Recall	F1	Time	Gen
DNN + SGD	0.817	0.055	0.571	0.101	59s	Gen 1
DNN + Adam	0.782	0.051	0.631	0.094	315s	Gen 1
LSTM	0.897	0.382	0.662	0.484	86s	Gen 2
Bidirectional LSTM	0.886	0.360	0.726	0.481	186s	Gen 2
GRU	0.899	0.390	0.683	0.496	22s	Gen 2
ClinicalBERT Frozen	0.860	0.148	0.250	0.186	1326s	Gen 3
ClinicalBERT Full FT	0.951	0.649	0.500	0.565	6239s	Gen 3

Table 5: Complete results sorted by generation

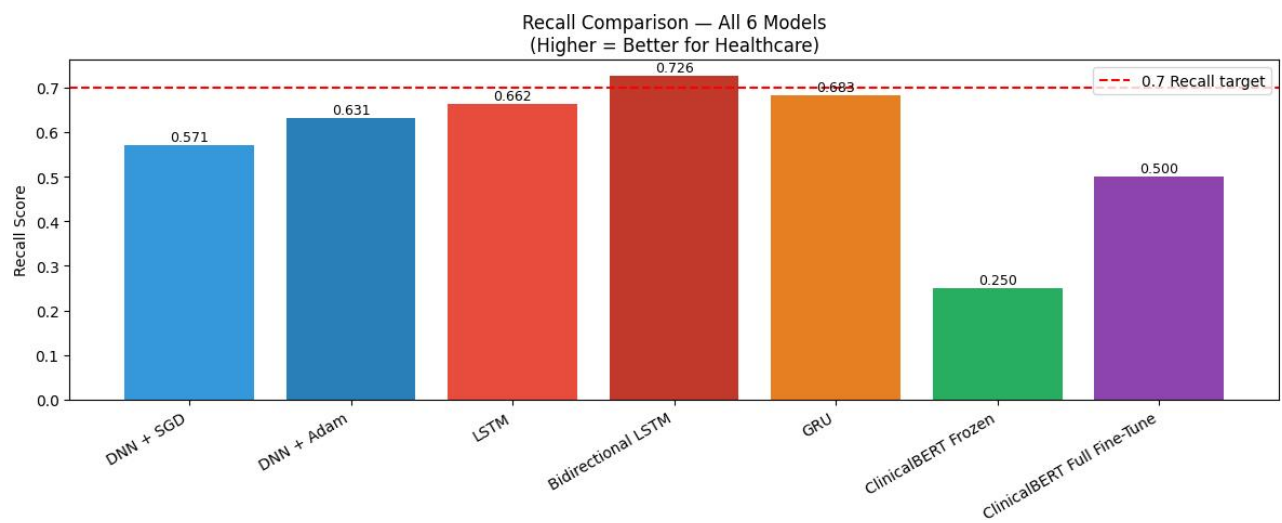


Figure 12: Recall Comparison All 6 Models (red dashed = 0.70 clinical target)

What this chart tells us: Only Bi-LSTM (0.726) crosses the 0.70 Recall target. GRU (0.683) and LSTM (0.662) are just below it. Both DNN models sit at 0.57-0.63 temporal context is the missing piece. ClinicalBERT Frozen is the worst at 0.25. Fine-tuned BERT at 0.50 is below target. Gen 2 models clearly dominate Recall. The +0.095 jump from DNN to Bi-LSTM confirms temporal patterns are the key driver.

Deployment: GRU recommended for live ICU monitoring. Recall 0.683, trains in 22s, no GPU needed, generates hourly alerts. Bi-LSTM (0.726) for offline analysis only needs future data. ClinicalBERT Full FT as secondary confirmation alongside GRU.

Ethics: GRU still misses 31.7% of sepsis patients. Must be a support tool, not a replacement. Fairness audits across demographics required before deployment. Clinicians must be able to override at any time.

X-Ray Extension: Adding chest X-rays creates a multimodal system: GRU branch for vitals, DenseNet-121 for imaging, ClinicalBERT for notes, all three fused before final classifier. Multimodal models consistently outperform single modalities. Key challenges: timestamp alignment and GPU compute requirements.

Summary: Gen 1 DNN : Recall 0.631 from flat vitals. Gen 2 time-series : Recall 0.726, the project's key breakthrough. GRU (Recall 0.683, 22s) recommended for live deployment. Gen 3 ClinicalBERT : 95% accuracy, best precision (0.649), interpretable attention. Best pipeline: GRU for real-time alerts + ClinicalBERT for confirmation.